Mirko Casu

AI Researcher · Licensed Clinical Psychologist

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RESEARCH INTERESTS

Language models and AI agents and their effects on mental health and human safety; bias in human and machine judgement; human perception of synthetic media; AI for clinical psychology.

EDUCATION

PhD, Computer Science · University of Catania · defence pending 2023 – 2026
Thesis: *Human Behaviour as Target and Medium of Artificial Intelligence in Clinical and Forensic Contexts*
Advisor Prof. Sebastiano Battiato · co-advisor Prof. Pasquale Caponnetto · IPLab

MSc, Clinical Psychology · University of Catania 2020 – 2023
110/110 cum laude · thesis on a smoking-cessation app, published in *BMC Psychology*

BSc, Psychological Sciences · University of Sassari 2015 – 2019
105/110 · thesis on video games, violence, rehabilitation and addiction

EXPERIENCE

AI Research Consultant · Eternal Tech, Acireale, Italy Sep 2026 – Present
Company building AI infrastructure for digital identity, persistent memory and intelligent agents

- Design the evaluation methodology and criteria for a proprietary AI technology, and build the benchmarking system that applies them
- Analyse the technology's architecture and write the scientific manuscript reporting its evaluation

Visiting Doctoral Researcher · MINTS Lab, EPFL, Lausanne, Switzerland Oct 2025 – Mar 2026
Host Prof. Andrea Cavallaro

- Ran a controlled comparison of **diffusion and autoregressive LLMs as automated judges**, characterising their evaluation biases; submitted to EACL 2027
- Analysed conversational trajectories between people and AI assistants

PhD Researcher · IPLab, University of Catania, Catania, Italy Aug 2023 – Jul 2026
SAMOTHRACE project · AI for wearables and cognitive bias in forensics

- Built **ORION**, an end-to-end pipeline that detects smoking episodes from wrist motion sensors by fusing 7 classifiers; on 40 participants (leave-one-subject-out) it reached an event-level F1 of 0.966 with no false positives in 29.5 hours of recordings (under review)
- Introduced the **impostor bias**, a systematic doubt that real media are authentic, and validated it with 746 participants whose accuracy at spotting AI images sat at chance (47–55.5%)
- Co-built two forensic benchmarks, **WILD** (20,000 images, 20 generators) and **SynthForensics** (6,815 videos, 5 text-to-video models; NeurIPS 2026), on which state-of-the-art detectors lose 29 AUC points on average
- Co-developed an **on-device autism screening** pipeline from children's gaze, distilled into a 1.6M-parameter ONNX ensemble that runs offline in an open-source Android app
- Led scoping reviews on AI chatbots for mental health (300+ citations, cited in UK Parliament POSTnote 737), wearables for smoking detection (49 studies, *IEEE JBHI*) and AI for eating disorders (7,545 records ranked by an LLM)

Early professional experience, Catania, Italy 2021 – 2023

- Specialist assistant for pupils with disabilities (I Girasoli cooperative); clinical trainee in applied psychoanalysis (APA Catania); videomaker for the University of Catania, ECLAT and CoEHAR

PUBLICATIONS

Own name in bold. Links lead to the DOI or to the paper.

Conference papers

SynthForensics: Benchmarking and Evaluating People-Centric Synthetic Video Deepfakes
R. Leotta, S. A. Sambataro, C. V. Ragaglia, **M. Casu**, Y. Petralia, F. Guarnera, L. Guarnera, S. Battiato · In *Advances in Neural Information Processing Systems (NeurIPS), Evaluations & Datasets Track*, 2026 · Accepted, to appear

Deepfake Forensic Analysis: Source Dataset Attribution and Legal Implications of Synthetic Media Manipulation
M. Cassia, L. Guarnera, **M. Casu**, I. Zangara, S. Battiato · In *2025 IEEE International Conference on Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRAINE)*, 2025

WILD: a new in-the-Wild Image Linkage Dataset for synthetic image attribution
P. Bongini, S. Mandelli, A. Montibeller, **M. Casu**, et al. (17 authors) · In *2025 International Joint Conference on Neural Networks (IJCNN)*, 2025

A Case Study on the Use of Hypnosis Combined with Virtual Reality in Psychological Partner Violence
M. Casu, S. Bellissima, G. Farruggio, C. Farrauto, P. Caponnetto · In *2024 IEEE International Conference on Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRAINE)*, 2024

Journal articles

Protocol-Aware, On-Device Per-Image ASD Classification on Saliency4ASD via Multi-Stream Distillation
M. O. Spata, **M. Casu**, F. Rundo, S. Battiato · *IEEE Open Journal of the Computer Society*, 2026

Fraud is not just rarity: A causal prototype attention approach to realistic synthetic oversampling
C. Giusti, L. Guarnera, **M. Casu**, S. Battiato · *Knowledge-Based Systems*, 2026

Digital technologies and artificial intelligence in eating disorders: A scoping review of prevention, screening, and treatment approaches
M. Casu, L. Marletta, C. V. Ragaglia, P. Caponnetto, S. Battiato · *Computers in Human Behavior Reports*, 2026

Smoking Detection and Cessation: An Updated Scoping Review of Digital and Mobile Health Technologies
M. Casu, F. Guarnera, G. R. M. L. Rosa, S. Battiato, P. Caponnetto, R. Polosa, R. Emma · *IEEE Journal of Biomedical and Health Informatics*, 2025

A (Mid)journey Through Reality: Assessing Accuracy, Impostor Bias, and Automation Bias in Human Detection of AI-Generated Images
M. Casu, L. Guarnera, I. Zangara, P. Caponnetto, S. Battiato · *Human Behavior and Emerging Technologies*, 2025
746 participants; open data and code

Breaking the Silence: Exploring Peritraumatic Distress and Negative Emotions in Male and Female Physical Domestic Violence Victims
P. Caponnetto, V. Lenzo, A. Sardella, G. C. Prezzavento, **M. Casu**, M. C. Quattropani · *Health Psychology Research*, 2024

Breathing tech: digital health innovations for managing asthma-related psychological dimensions
M. Casu, P. Caponnetto · *Exploration of Digital Health Technologies*, 2024

A Multi-Level Analysis of Biological, Social, and Psychological Determinants of Substance Use Disorder and Co-Occurring Mental Health Outcomes
C. I. Belfiore, V. Galofaro, D. Cotroneo, A. Lopis, I. Tringali, V. Denaro, **M. Casu** · *Psychoactives*, 2024

Psychological Factors, Digital Health Technologies, and Best Asthma Management as Three Fundamental Components in Modern Care: A Narrative Review
P. Caponnetto, G. C. Prezzavento, **M. Casu**, R. Polosa, M. C. Quattropani · *Applied Sciences*, 2024

Behavioral Therapy for People With Diabetes Who Smoke: A Scoping Review
R. Sammut, J. Grech, R. Polosa, ..., **M. Casu**, et al. (17 authors) · *Journal of Primary Care & Community Health*, 2024

Exploring the Therapeutic Potential of Virtual Reality: A Review on the Simulation of Psychedelic Effects for Treating Psychological Disorders
M. Casu, C. Farrauto, G. Farruggio, S. Bellissima, S. Battiato, P. Caponnetto · *Psychology International*, 2024

Night Eating Syndrome: A Review of Etiology, Assessment, and Suggestions for Clinical Treatment
M. Bargagna, **M. Casu** · *Psychiatry International*, 2024

GenAI mirage: The impostor bias and the deepfake detection challenge in the era of artificial illusions
M. Casu, L. Guarnera, P. Caponnetto, S. Battiato · *Forensic Science International: Digital Investigation*, 2024
Introduced the impostor bias; referenced by INTERPOL (Project SynthWave)

AI Chatbots for Mental Health: A Scoping Review of Effectiveness, Feasibility, and Applications
M. Casu, S. Triscari, S. Battiato, L. Guarnera, P. Caponnetto · *Applied Sciences*, 2024
Cited in UK Parliament POSTnote 737; most-cited paper (300+ citations)

Rolling the Dice: A Comprehensive Review of the New Forms of Gambling and Psychological Clinical Recommendations
M. Casu, C. I. Belfiore, P. Caponnetto · *Psychiatry International*, 2023

User evaluation and feasibility test of an app designed for smoking cessation in Italian people who smoke: preliminary findings from an uncontrolled pre-test post-test open study
P. Caponnetto, **M. Casu**, D. Crane, L. Ross, M. C. Quattropani, R. Polosa · *BMC Psychology*, 2023

Update on cyber health psychology: Virtual reality and mobile health tools in psychotherapy, clinical rehabilitation, and addiction treatment
P. Caponnetto, **M. Casu** · *International Journal of Environmental Research and Public Health*, 2022

New Perspectives in Psychopathology and Psychological Well-being by Using Forest Therapy: A Systematic Review
P. Caponnetto, L. Inguscio, S. Triscari, **M. Casu**, A. Ferrante, D. Cocuzza, M. M. Maglia · *The Open Psychology Journal*, 2022

The effects of physical exercise on mental health: From cognitive improvements to risk of addiction
P. Caponnetto, **M. Casu**, M. Amato, et al. (11 authors) · *International Journal of Environmental Research and Public Health*, 2021

Book chapters

Sistemi di IA ad alto rischio di cui all'Articolo 6, Paragrafo 2
M. Casu, A. Di Pierno, O. Pontorno, I. Zangara, L. Guarnera, S. Battiato, D. Curtotti · In *AI Act e Legge nazionale sull'Intelligenza Artificiale*, Giuffrè, 2026

Workshop and challenge papers

Causal Prototype Attention Classifier: An Interpretable Model for Credit Card Fraud Detection Under Extreme Class Imbalance
C. Giusti, **M. Casu**, L. Guarnera, S. Battiato · In *Joint Proceedings of the Thematic Workshops at Ital-IA 2025*, 2025

(DFF '25) 1st Deepfake Forensics Workshop: Detection, Attribution, Recognition, and Adversarial Challenges in the Era of AI-Generated Media
S. Battiato, **M. Casu**, F. Guarnera, L. Guarnera, G. Puglisi, O. Pontorno, C. V. Ragaglia, Z. Akhtar · In *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025

Adversarial Attacks on Deepfake Detectors: A Challenge in the Era of AI-Generated Media (AADD-2025)
S. Battiato, **M. Casu**, F. Guarnera, L. Guarnera, G. Puglisi, O. Pontorno, C. V. Ragaglia, Z. Akhtar · In *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025

Under review

ORION, smoking-episode detection from wrist motion sensors
Manuscript under review

Evaluation bias of diffusion and autoregressive LLMs used as automated judges
Submitted to EACL 2027 · title withheld during double-blind review

Preprints

Virtual Reality Neurofeedback Training combined with Trauma-Informed Motivational Interviewing for the Treatment of Post-Traumatic Stress Disorder
S. Triscari, **M. Casu**, E. Uccelli, N. M. Vitale, V. Rapisarda, A. Fakhrou, P. Caponnetto · *Preprints.org*, 2024

The “Slotstreams” Phenomenon on Twitch. Tv: Can it Lead to Online Gambling?
M. Casu, C. I. Belfiore · *Preprints.org*, 2022

OPEN MODELS, DATA AND SOFTWARE

Apertus 1.5 8B for Apple silicon: 12 MLX quantizations (text and vision) of the Swiss open LLM, benchmarked for perplexity and psychological profile (PsychoBench); 1,900+ downloads in the last month — 2026

LLaDA 2 diffusion LLMs: 4 quantizations of LLaDA 2.1-flash and 2.2-mini, in MLX for Apple silicon and NVFP4 for NVIDIA GPUs — 2026

ASD Screening Android app: Open-source app, ONNX models and training code for the IEEE OJCS paper (Apache-2.0) — 2026

A (Mid)journey dataset: 75 stimuli and 11,170 human judgements, with Grad-CAM maps — 2025

SERVICE

Session Chair, Technical Session 6, IEEE ICAISF 2026, Catania — 2026

Track Co-Chair, DFF-2025 Deepfake Forensics Workshop, ACM Multimedia 2025 — 2025

Dataset Committee, AADD-2025 challenge (13 teams), ACM Multimedia 2025 — 2025

Lead Organizer, FHDA challenge, IEEE MetroXRAINE 2025 — 2025

Guest Editor (2023–25) and **Academic Editor** (2024–25), special issue of *Exploration of Digital Health Technologies* (4 articles) — 2023 – 2025

Reviewer for MDPI journals (*Healthcare*, *IJERPH*, *Psychiatry International*) — 2023 – Present

TALKS AND POSTERS

Psychedelic simulation in virtual reality, DigiTales webinar, host Prof. Steffen Moritz — 2025

WILD dataset, VERIMEDIA Workshop at IJCNN 2025, Rome — 2025

Digital and mobile health for smoking detection, Global Forum on Nicotine, Warsaw — 2024

Smartwatch-based dual-diagnosis project, AIP Clinical and Dynamic Psychology Congress — 2024

AI for smoke-free living through wearables, European Conference of Digital Psychology — 2024

Impostor bias (poster), Psicologia in Cantiere, Palermo — 2024

TEACHING AND PUBLIC ENGAGEMENT

Speaker and staff, IFOSS Forensics Summer School (3 editions) — 2024 – 2026

We Make Future, Bologna · demonstrator for the University of Catania and CoEHAR — 2025

SHARPER Night, Catania · virtual reality and wearable phobia-response demo — 2024

Didacta Italia, Catania · for the Sicilian Order of Psychologists — 2023

LICENSURE

Licensed Psychologist · Order of Psychologists of Sicily, Section A, reg. no. 12260 — since Sep 2024

SKILLS AND LANGUAGES

Machine learning: Python, PyTorch, Hugging Face, MLX, scikit-learn, ONNX Runtime, LLM evaluation, quantization (MLX, GGUF, NVFP4), knowledge distillation, explainability (Grad-CAM)

Human research: experimental design, mixed methods, questionnaire design, inferential statistics (Jamovi), PRISMA scoping reviews, LLM-assisted screening

Clinical and devices: psychological assessment, virtual reality, wearables, mHealth, Android

Languages: Italian (native) · English (C1, IELTS Academic 7.0) · French (basic)

Last updated October 2026 · full publication list with PDFs and code at mirkocasu.github.io/publications